

CCNA

VLAN and INTER-VLAN Routing TASK – 6

1. On a Cisco Layer 3 switch, create three VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support', VLAN 30 for 'Marketing') and assign at least one interface to each VLAN.

Ans :

- enable
- configure terminal
-
- vlan 10
- name Sales
- exit
-
- vlan 20
- name Support
- exit
-
- vlan 30
- name Marketing
- exit
-
- interface fastEthernet 0/1
- switchport mode access
- switchport access vlan 10
- no shutdown
- exit
-
- interface fastEthernet 0/2

- switchport mode access
- switchport access vlan 20
- no shutdown
- exit
-
- interface fastEthernet 0/3
- switchport mode access
- switchport access vlan 30
- no shutdown
- exit
-
- end
- show vlan brief

**2. Configure Switch Virtual Interfaces (SVIs) for each VLAN you created by assigning them unique IP addresses (e.g., 192.168.10.1/24 for VLAN 10, etc.) and ensure each SVI is in 'up/up' state.

Hint: Use the 'interface vlan' command to create SVIs and 'no shutdown' to enable them.**

Ans :

- enable
- configure terminal
-
- ip routing
-
- interface vlan 10
- ip address 192.168.10.1 255.255.255.0
- no shutdown
- exit
-

- interface vlan 20
- ip address 192.168.20.1 255.255.255.0
- no shutdown
- exit
-
- interface vlan 30
- ip address 192.168.30.1 255.255.255.0
- no shutdown
- exit
-
- end
- show ip interface brief

**3. Enable IP routing on the Layer 3 switch so that devices in different VLANs can communicate with each other without using an external router.

Hint: Use the 'ip routing' global configuration command.**

Ans :

- enable
- configure terminal
- ip routing
- end
- show running-config | include ip routing
- show ip interface brief

4. Verify inter-VLAN routing by connecting two PCs (one in VLAN 10 and one in VLAN 20) and pinging between them. Capture and submit the ping results to show successful communication.

Ans :

- enable

- show ip interface brief
- show ip route
- ping 192.168.20.10

5. Refactor your VLAN and SVI configuration so that VLAN 30 uses a different subnet mask (e.g., /25 instead of /24). Explain what changes you made and how this affects communication with the other VLANs.

Ans :

- enable
- configure terminal
- interface vlan 30
- ip address 192.168.30.1 255.255.255.128
- no shutdown
- exit
- end
- show ip interface brief
- show ip route

VLAN 30 can still communicate with VLAN 10 and VLAN 20 through the Layer 3 switch because the VLANs are **different IP networks** and ip routing is enabled. However, all VLAN 30 devices must use the new /25 mask and stay within the 192.168.30.0–192.168.30.127 subnet. Devices configured with the old /24 mask may experience incorrect subnet calculations and connectivity problems